# Carbon and its Compounds

## The big idea

Carbon forms millions of compounds: fuels, plastics, medicines and the molecules of life. It does this because of **covalent bonding**, **catenation** and **tetravalency**.

## Covalent bonds

Carbon (atomic number 6, configuration 2, 4) can neither gain 4 nor lose 4 electrons easily, so it **shares** electrons. A **covalent bond** is a shared pair of electrons.

Examples: $H_2$ (single bond), $O_2$ (double bond), $N_2$ (triple bond), $CH_4$ (four single bonds). Covalent compounds have **low melting and boiling points** and are **poor conductors** of electricity.

## Why carbon is special

**VERSATILITY OF CARBON**

**Catenation**: carbon atoms link with other carbon atoms to form long chains, branched chains and rings.
**Tetravalency**: a valency of 4 lets carbon bond with four other atoms (C, H, O, N, S, Cl…).
Carbon–carbon bonds are very strong, making these compounds stable.

**Allotropes** of carbon: diamond (each C bonded to 4 others; hardest), graphite (layers of hexagons; conducts electricity, slippery), fullerene $C_{60}$ (football shape).

## Saturated and unsaturated compounds

- **Saturated** (single bonds only): alkanes, $C_nH_{2n+2}$, e.g. ethane $C_2H_6$.
- **Unsaturated** (double or triple bonds): alkenes $C_nH_{2n}$ (ethene $C_2H_4$), alkynes $C_nH_{2n-2}$ (ethyne $C_2H_2$).

**Homologous series**: compounds with the same functional group whose successive members differ by **$-CH_2-$** (14 u). They have similar chemical properties and gradually changing physical properties.

| Functional group | Suffix/prefix | Example |
|---|---|---|
| Halogen (–Cl, –Br) | chloro-, bromo- | chloropropane |
| Alcohol (–OH) | -ol | propanol |
| Aldehyde (–CHO) | -al | propanal |
| Ketone (>C=O) | -one | propanone |
| Carboxylic acid (–COOH) | -oic acid | propanoic acid |

## Chemical properties

- **Combustion**: carbon compounds burn in oxygen giving $CO_2$, $H_2O$, heat and light. Saturated compounds burn with a clean blue flame; unsaturated ones with a sooty yellow flame.
- **Oxidation**: alcohols → carboxylic acids using alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$.
- **Addition**: unsaturated + $H_2$ (Ni/Pd catalyst) → saturated. Used in **hydrogenation of vegetable oils**.
- **Substitution**: saturated hydrocarbons + $Cl_2$ in sunlight; H atoms are replaced by Cl.

## Ethanol and ethanoic acid

- **Ethanol** ($C_2H_5OH$): the active ingredient of alcoholic drinks and a common solvent. Reacts with sodium to give hydrogen; with hot concentrated $H_2SO_4$ it dehydrates to ethene.
- **Ethanoic acid** ($CH_3COOH$): 5–8% solution is **vinegar**; pure acid freezes in winter (glacial acetic acid). Turns blue litmus red.

**ESTERIFICATION**

Ethanoic acid + ethanol → (acid catalyst) **ester** (sweet-smelling, used in perfumes) + water.
**Saponification**: ester + NaOH → alcohol + sodium salt of the acid (soap).

## Soaps and detergents

A soap molecule has an **ionic (hydrophilic) end** that dissolves in water and a **long hydrocarbon (hydrophobic) tail** that dissolves in oil. In water, soap molecules form **micelles**, trapping oily dirt in the centre so it can be rinsed away.

Soap forms **scum** with the calcium and magnesium salts of hard water; **detergents** don't, so they work in hard water.

## Quick recap

- Covalent bonding, catenation and tetravalency make carbon versatile.
- Homologous series differ by $-CH_2-$; functional groups decide properties.
- Combustion, oxidation, addition, substitution.
- Micelles explain how soap cleans.

## Practice questions

1. The electronic configuration of carbon is
   (a) 2, 6 (b) 2, 4 (c) 2, 8 (d) 4, 2
2. The property of carbon to form long chains is called
   (a) tetravalency (b) catenation (c) isomerism (d) allotropy
3. Successive members of a homologous series differ by
   (a) CH (b) $CH_2$ (c) $CH_3$ (d) $C_2H_4$
4. Hydrogenation of vegetable oils uses a catalyst such as
   (a) iron (b) nickel (c) copper (d) platinum only
5. Soaps do not work well in hard water because they
   (a) form scum with calcium and magnesium ions (b) dissolve too quickly (c) are acidic (d) form micelles
6. The functional group of an aldehyde is
   (a) –OH (b) –CHO (c) –COOH (d) >C=O

Answers: 1. b 2. b 3. b 4. b 5. a 6. b